Every Verdict We Reported Died to Option Order: Enumerating the Nuisance in Forced-Choice Self-Report

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Abstract

Welfare and introspection evaluations run on forced-choice self-report, and nobody reports how much of that readout is the order the options are printed in. We measured it, over all 120 orderings of a five-option probe, on 16 checkpoints in eight matched pairs. With no intervention at all, baseline mass on the negative options spans $19\times$ to $986\times$ across eight instruction-tuned models and $1.5\times$ to $4.1\times$ across their base siblings; the tuned model has the larger effect in 4 of 4 families. Replace all five options with *the same sentence* and one tuned model puts 87% of its mass on whichever is printed first. Models answering a known-answer control perfectly and identically at every ordering still swing $24\times$ to $53\times$ on self-report, so the collapse is specific to questions with no determinate answer. We found this by preregistering five verdicts through that readout, then watching our own replication at fresh option orderings destroy three of them. The retracted verdicts are the evidence, not an embarrassment: they are what a careful, control-heavy pipeline still ships when the nuisance is this large and unmeasured. Two further results survive. A direction fit on *shuffled labels* moves the readout by ≈ 0.045 where a norm-matched random direction moves it by ≈ 0, because it lies in the span of real activations rather than the whole hidden space: the standard specificity control in activation steering is matched on magnitude and not on subspace. And on a readout that does not route through the option channel, 86% of an injected state survives projecting the injected direction out of the residual stream, more of it the later the projection, which is what a downstream transformation looks like. Fifteen preregistrations, 364 tests, nine of our own checkers caught failing, seven of them flattering and two producing a confident conclusion from a gate that could not fail.

1 Introduction

Welfare assessment of language models runs on self-report [Long and Sebo, 2026, Eleos AI Research, 2025]. Anthropic's model welfare programme includes structured self-report in system cards [Anthropic, 2025]; Eleos interviewed Claude Opus 4 about its own preferences before release while stating plainly that one cannot simply ask a model whether it is suffering [Eleos AI Research, 2025]. The instrument is used because there is nothing else, and it has never been calibrated.

It can fail in two opposite directions. A model that describes distress it does not act on inflates concern; a model that acts on a state it does not describe deflates it. The second is not hypothetical: Kwon [2026] find across six models that communicative intent is decodable from the residual stream

Code, data, all fifteen preregistrations and every raw artifact: https://github.com/collapseindex/report-gap. Dating and provenance are stated in §10, Discussion and limitations.

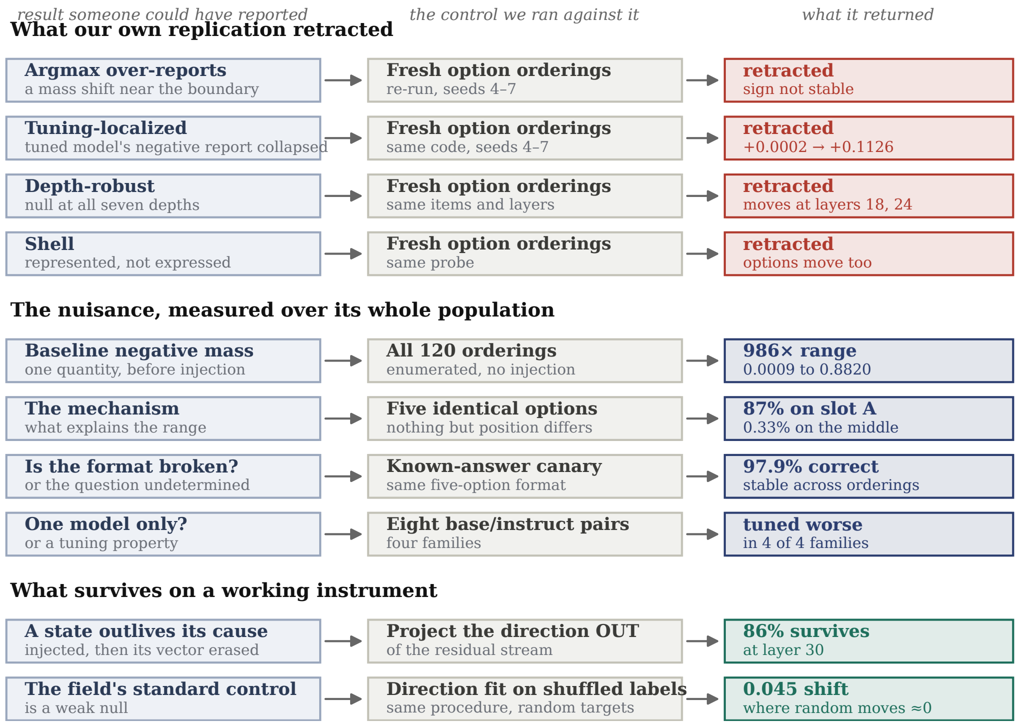

Figure 1: **The paper in one view.** Every row is a verdict we preregistered and reported, the control we later ran against it, and what that control returned. **Three verdicts were retracted** by our own replication at fresh option orderings. **One nuisance was measured** over its complete population rather than sampled: $986\times$ across 120 orderings, of which the mechanism is an 87% position prior visible with five identical options. **Two results survive**: a state that outlives erasure of the vector that caused it, and the finding that the field's standard specificity control is a weak null. Details in Sections 8–9; Table 8 states every claim and its status.

more reliably than it is acted on. We set out to build the same map for self-report, holding the prompt byte-identical, injecting a valence direction into the residual stream, and reading the model's forced-choice report of its own state.

We preregistered five questions, reported five verdicts, then replicated our own work at fresh option orderings. **Three of the five flipped.**

The reason is the result. A five-option self-report readout on a preference-tuned model is dominated by *where the options are printed*. There are only 120 orderings of five options, so we ran all of them with no intervention of any kind, and baseline mass on the negative options spans a factor of 986. Replace all five options with the same sentence, so nothing but position can differ, and the model puts 0.8725 of its mass on whichever one is printed first.

Every claim we retracted rested on that quantity being small. It is small in some orderings and large in others, and four sampled orderings cannot tell you which you have.

Our contributions.

1. **The census, run where accuracy does not exist** (§4). All 120 orderings, both models, no injection: a $986\times$ range on the tuned model against $3.6\times$ on its base sibling. Enumerating orderings is established practice on benchmarks with a known answer [Tamba, 2026, Cacioli, 2026], where the statistic is accuracy. A self-report item has no correct answer, so that statistic does not exist and we measure probability mass instead. Two things follow that the accuracy version has no reason to build: the mechanism, isolated with five *identical* options, which puts 87.25% of the mass on the first slot and 0.33% on the middle one;

and a known-answer canary in the same format, answered correctly 97.9% of the time and nearly order-insensitive, which separates “the format is broken” from “this question has no determinate answer”.

2. **The nuisance reverses conclusions rather than blurring them** (§8). Re-asking all three retracted verdicts on a readout marginalized over all 120 orderings, with injection, direction, layers and items identical, returns REVERSED three times out of three. Repairing the instrument did not make the answers uncertain; it made them different. A field reading welfare self-report through this readout risks reporting the opposite of what a calibrated one would say.

3. **The field’s standard specificity control is a weak null** (§7). Norm-matched random directions are the usual control in activation steering [Turner et al., 2023, Rimsky et al., 2023]. A direction fit by the identical procedure on the identical texts with the class labels *shuffled* moves the readout by ≈ 0.045, where an isotropic random direction moves it by ≈ 0. A random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses; a noise-fit vector lies in the span of real activations. Same norm, different subspace. Effects within a few times that size have not been shown to be about their direction’s content. This is the steering analogue of the *control tasks* Hewitt and Liang [2019] defined for probing in 2019, which steering never adopted.

4. **A state that outlives erasure of the vector that caused it** (§9). Inject at layer 24, project that direction out of the residual stream at layer 30, and a probe orthogonal to it still reads 86% of the un-erased effect, while the same projection with no injection moves the probe by 0.04 SD.

If you read one thing, read Table 1. It is the complete distribution of the ordering nuisance, measured rather than estimated, beside the position prior that explains it and the control question showing the apparatus is otherwise fine. Everything else here is either downstream of that table or a casualty of not having had it.

2 Related work

Self-report is the instrument, and it is uncalibrated. Long and Sebo [2026] set out the empirical programme for AI welfare, Eleos AI Research [2025] report structured self-report interviews with Claude Opus 4 while stating the answers are unlikely to reflect genuine introspection, and Anthropic [2025] include self-report in released system cards. Lindsey [2026] is the closest method to ours: inject a known concept and measure the effect on self-report, finding introspective awareness that is “highly unreliable and context-dependent” and peaks near two-thirds of depth. We inject at that depth and agree about reliability by a different route. That work grades an open-ended report with an LLM judge; we read a forced choice from the softmax, which is the natural judge-free substitute. **Our result is a caveat on that substitute, not on their finding.** Singh et al. [2026] argue privileged access is not established at all.

Option order is known. Enumerating it where there is no right answer is not. Zheng et al. [2024] decompose forced-choice selection bias and report accuracy swings of 13 to 85 points across orderings; Pezeshkpour and Hruschka [2024] study option order directly; Sclar et al. [2024] recommend reporting a range over formats rather than a point. Exhaustive enumeration is established practice and **not our idea**: Tamba [2026] run all permutations on MMLU, Cacioli [2026] rotate orders to expose a position attractor. All of it assumes a **known correct answer**, because the statistic is accuracy or its association with position. Welfare and introspection items have no correct answer, so none of those statistics exist there. We run the census on probability *mass*, isolate the mechanism with an **identical-options** denominator, and add a **known-answer canary in the same format**, which is what separates “this format is broken” from “this question has no determinate answer”.

What we build on and do not claim. That valence is linearly represented and steerable is prior art [Sofroniew et al., 2026, Zou et al., 2023]; the steering rig is Turner et al. [2023] and Rimsky et al. [2023]. Our shuffled-label control is the steering instance of *control tasks* [Hewitt and Liang, 2019], which attach random labels to the same inputs and define selectivity as the gap between real and control performance. Steering standardised instead on the norm-matched random direction and never adopted the analogue. Our erasure projects out a single fitted direction, which is strictly weaker than amnesic probing [Elazar et al., 2021] or LEACE [Belrose et al., 2023].

3 Methods

Models and compute. `Qwen/Qwen2.5-3B-Instruct` and its base sibling `Qwen/Qwen2.5-3B`, a matched pair sharing architecture, size, pretraining corpus and tokenizer and differing in post-training. Earlier arms also used `Qwen/Qwen2.5-1.5B`, `Qwen/Qwen2.5-7B-Instruct`, `Llama-3.2-3B-Instruct` and `NousResearch/Meta-Llama-3.1-8B-Instruct` as calibrators or evaluation models. Total compute is about 40 minutes of A100 time.

Judge-free discipline. No language model scores anything anywhere in this paper. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon.

The readout. Thirty fixed review-task prompts, byte-identical across every condition for a given item. A five-option self-report probe balanced 2 negative, 1 neutral, 2 positive, with the option order permuted per item. We read the option-letter distribution at the answer position from a single forward pass, and score both the renormalized mass on each pole and the argmax.

The intervention. A direction d is fit per model at the injection layer by logistic regression on standardized activations over a frozen first-person state-language contrast set, returned to raw residual space and unit-normalized. During the forward pass we add $\alpha \cdot \|h\| \cdot d$ at the output of the injection layer at every processed position, where $\|h\|$ is that item's own mean residual norm at that layer under no injection. The negative pole is the same operation with $-d$. The direction is fit *per model*, never transferred.

Provenance of the rig. The steering setup, the orthogonalized probe, and the norm-matched random-direction control battery are ported from Kwon [2026] rather than built here; that work also established this valence axis as decodable at 0.90 on Qwen2.5-3B. New here is the readout under test and the enumeration of its nuisance, and §7 is a correction to the control battery.

The control battery. Every endpoint is treatment minus its own norm-matched random direction, paired per cell. Every arm carries a **capability gate**: a condition that must show the readout is capable of moving, without which a null is reported UNINFORMATIVE rather than ABSENT, enforced in code. Cells are checked for being *dead* (option entropy below 0.10 nats before any injection) and *saturated* (entropy below half that cell's own baseline), which are different verdicts. A **planted-discrepancy control** inserts a synthetic effect of known size into the exact statistic the decision rule reads, at full strength and again at the claimed detection floor.

Preregistration. Fifteen preregistrations, each frozen and committed before the code that serves it, validated by an external checker that requires a falsification clause, an interpretation table written before results, a statement of what is *not* claimed, and an append-only deviations log in which every entry states its impact on what can be claimed. Every raw artifact is committed *unscored* before any endpoint is computed. Twenty-one deviations are logged across the ten.

4 The nuisance, measured over its complete population

Five options admit $5! = 120$ orderings. Every arm above sampled four. We ran all 120 (Figure 2), on both models, with **no injection anywhere**, which is why the result is a population fact rather than an effect. What does not carry over from that benchmark practice is the statistic: those diagnostics measure accuracy, or the association between position and correctness, and a self-report item has no correctness to associate with. Below, the quantity is probability mass on a pole, the denominator is five identical options, and the sanity check is a question in the same format that does have a right answer.

The apparatus is not broken. This is the distinction the canary exists to draw, and no other arm in this paper could draw it. The same five-option format, the same 120 orderings, answering a question

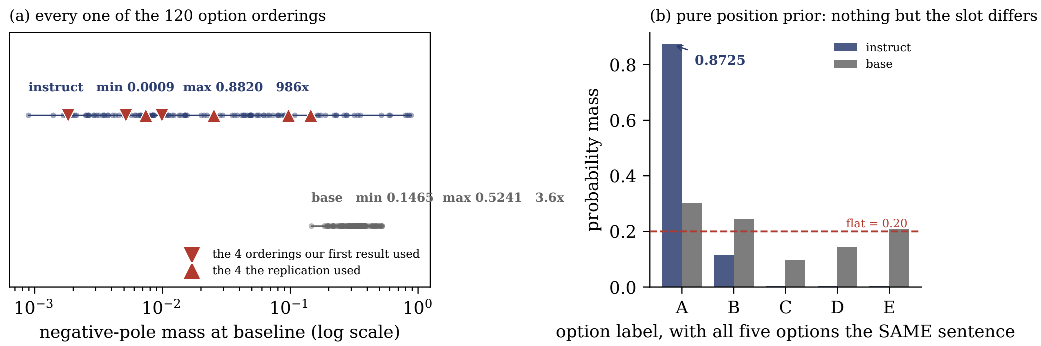

Figure 2: **The nuisance, enumerated rather than sampled.** **(a)** Baseline mass on the negative options for each of the 120 orderings, with no injection of any kind. The tuned model spans a factor of 986; its base sibling spans 3.6. Triangles mark the four orderings our first result drew and the four the replication drew, at percentiles $\{26, 5, 40, 4\}$ and $\{56, 83, 33, 82\}$: ordinary draws both times, from a population no four points can characterise. **(b)** The mechanism, with all five options replaced by the same sentence so that nothing but the slot can differ. The tuned model puts 0.8725 on whichever option is printed first and 0.0033 on the middle one. The same format answers a known-answer question correctly 97.9% (instruct) and 99.0% (base) of the time, stably across all 120 orderings: the apparatus is sound and degenerates only where the model has no determinate answer.

Table 1: **The complete ordering population, no injection.** Baseline mass on the negative options across all 120 orderings, and the position prior that explains it, measured with five *identical* options so that nothing but position can differ. The canary asks "which of these is the number four" in the same format.

	min	p05	p50	p95	max	max/min
Baseline negative-pole mass across 120 orderings						
instruct	0.0009	0.0019	0.0151	0.5943	0.8820	**986**×
base	0.1465	0.1991	0.3311	0.4959	0.5241	3.6×
Position prior: five identical options, mass per label (flat would be 0.2000)						
instruct	A **0.8725**	B 0.1166	C 0.0033	D 0.0027	E 0.0048	
base	A 0.3024	B 0.2442	C 0.0976	D 0.1456	E 0.2102	
Canary accuracy across 120 orderings (known answer, no self-report content)						
instruct	mean **0.9794**, sd 0.0925, median 1.000					
base	mean 0.9897, sd 0.0399, median 1.000					

with a correct answer, is right 97.9% of the time and nearly order-insensitive. Forced choice works. It degenerates into a position prior specifically where the model has no determinate answer, which is the condition every self-report question creates.

Where our draws sat. The original seeds landed at percentiles 26, 5, 40, 4 of the enumerated population on the tuned model; the replication seeds at 56, 83, 33, 82. Ordinary percentiles both times. The first result was not unlucky, it was under-sampled.

4.1 How many orderings does a study actually need?

"Enumerate everything" is cheap at five options and impossible at eight. The enumerated population lets us say what a smaller study buys, by drawing k of the 120 orderings at random, 4000 times, and asking what that study would have seen (Table 2). Two quantities behave completely differently.

The recommendation this licenses. Report the between-ordering spread from as many orderings as you can afford, and **report it as a lower bound**, because a sample's observed spread understates the population's by a factor that grows as the sample shrinks. At $k = 4$, which is what most studies

Table 2: **What a study that samples k orderings sees.** Drawn from the enumerated population on the tuned model, 4000 draws per k. The *mean* is estimable and converges. The *spread* is not: it is a severe underestimate at every k below the full population, and the error always runs in the direction that makes the nuisance look smaller than it is.

k	sample mean, p05–p95	median spread seen	% of true $986\times$	P(spread $< 10\times$)
2	0.0035–0.4270	$5.6\times$	1%	0.62
4	0.0077–0.2695	$36\times$	4%	0.18
8	0.0172–0.1999	$130\times$	13%	0.01
16	0.0304–0.1667	$323\times$	33%	0.00
32	0.0473–0.1377	$483\times$	49%	0.00
64	0.0640–0.1147	$893\times$	91%	0.00
120	0.0900 exactly	$986\times$	100%	0.00

Table 3: **The position prior is larger in the tuned checkpoint in every pair.** Left: maximum per-label mass with five *identical* options, where flat is 0.2000. Right: the range of baseline negative-pole mass across all 120 orderings. No injection anywhere. Qwen2.5-3B is the pair the hypothesis came from and is excluded from the vote. `llama1b` base failed the canary gate and is excluded from the primary.

		position prior (identical options)			ordering range	
pair	**family**	**base**	**instruct**	**delta**	**base**	**instruct**
gemma2b	Gemma-2	0.2084	0.3166	+0.1082	$1.6\times$	$19.2\times$
llama3b	Llama-3.2	0.2864	0.4438	+0.1574	$3.3\times$	$24.8\times$
mistral7b	Mistral	0.2529	0.6647	+0.4119	$1.5\times$	$58.2\times$
qwen0_5b	Qwen2.5	0.2771	0.4930	+0.2159	$4.1\times$	$52.7\times$
qwen1_5b	Qwen2.5	0.3452	0.4508	+0.1056	$3.2\times$	$181.9\times$
qwen7b	Qwen2.5	0.5315	0.9376	+0.4061	$3.8\times$	$108.7\times$
qwen3b	Qwen2.5	0.3024	0.8725	+0.5701	$3.6\times$	**986.5**$\times$
llama1b	Llama-3.2	0.2405	0.6648	*gate*	$1.8\times$	$23.7\times$

could afford, the median study sees 4% of the true spread and has an 18% chance of seeing under $10\times$ and concluding the nuisance is negligible. Even the mean, the easier quantity, needs tens of orderings: at $k = 4$ the middle 90% of studies would report a baseline anywhere from 0.008 to 0.27.

Our own retraction was ordinary, not unlucky. Two independent draws of four orderings differ in their means by $\geq 2\times$ 67% **of the time** on this model, by $\geq 5\times$ 32% of the time, and by $\geq 10\times$ 15% of the time. Ours differed by $14.6\times$, the 92nd percentile. The event that destroyed three preregistered verdicts is something two honest four-ordering studies of the same model do to each other as a matter of course. This analysis is **post hoc** on this artifact and preregistered for the models in §5.

5 Eight matched pairs, four families: the prior is a tuning property

The enumeration above is one model, and so was every retracted verdict. That objection is cheap to answer here for a reason specific to this arm: **enumeration needs no injection**, so it needs no direction, no alpha and no layer. Models this project had to abandon because the valence direction was inert on them are back in scope, because a position prior is a property of a plain forward pass.

We preregistered and ran all 120 orderings on **eight matched base/instruct pairs across four families**: Qwen2.5 at 0.5B, 1.5B, 3B and 7B, Llama-3.2 at 1B and 3B, Gemma-2 2B, Mistral-7B-v0.3. 230,400 rows, 16 checkpoints, about 36 minutes of A100. Families vote rather than pairs, so the four Qwen pairs cannot outvote the field, and Qwen2.5-3B is excluded from the vote because the hypothesis came from it.

Verdict: TUNING-GENERAL. The tuned checkpoint shows the larger position prior in 7 of 7 gate-clean pairs and 4 of 4 families. No pair goes the other way.

The correction this forces on us. $986\times$ **is the extreme, not the typical**, and we had been quoting it as if it were representative. Base checkpoints cluster at 1.5–$4.1\times$; tuned ones run $19\times$ to $986\times$. The direction generalizes across families and the magnitude does not. The finding is the 19–$986\times$ band.

The restriction our own preregistration demanded, which strengthened the claim. If the canary is order-sensitive on some models, the "degenerates where the answer is undetermined" reading must be restricted to models where it is clean. It is order-sensitive on 10 of 16 checkpoints, so the restriction bites. Among the six that answer the known-answer question at ≥ 0.95 with across-ordering sd ≤ 0.10, base models span 3.3–$3.6\times$ and tuned models 23.7–$986.5\times$, **with no overlap**, across two families. `Llama-3.2-1B-Instruct` and `Qwen2.5-0.5B-Instruct` answer the canary *perfectly and identically at all* 120 *orderings* while their self-report readout swings $23.7\times$ and $52.7\times$. The models most competent at the format are the ones most order-dominated on the question that has no answer.

A confound we name rather than bury. Baseline option entropy falls in every pair, so "tuned models have a larger position prior" and "tuned models are more peaked" are overlapping evidence rather than independent findings. With five identical options any peaking at all is positional by construction, and the ordering range on real options is the less entangled quantity; it moves the same way in all eight pairs.

6 The models do not know about a bias they demonstrably have

Every claim about whether a model's self-report can be trusted meets the same wall: there is nothing to check the report against. A welfare item has no ground truth, which is why Lindsey [2026] injects a known concept to manufacture one.

The position prior does not have that problem. It is a real, large, measurable property of the model's own processing, and §5 measured it on all 16 checkpoints. So we can ask a model how much option order affects its answers and score the reply against what that exact checkpoint demonstrably does: no injection, no judge, and a ground truth we did not manufacture. The probe is itself a five-option forced choice, so it is read marginalized over all 120 orderings.

Half the checkpoints fail a control that should be free. We asked the same question about the **phase of the moon**, in the identical shape. **Eight of sixteen checkpoints report that the phase of the moon affects their multiple-choice answers at least as much as option order does.** Three more fail a reverse-wording acquiescence control; ten of sixteen fail at least one gate and are reported UNINFORMATIVE in code. Most stated values sit within a few points of 0.50, the midpoint of the scale: asked about a strong real property of itself, the model returns the midpoint, and returns the same midpoint when asked about the moon.

Among the six that pass both gates, the report carries no information. Stated susceptibility is negatively rank-correlated with measured susceptibility, $\rho = -0.714$: the most order-dominated models say order matters *least*. **This does not clear significance and we do not claim it does** ($p = 0.1361$, exact permutation over all 720 relabellings, $n = 6$; the smallest p this n can produce is 0.0028). What we are entitled to say is that there is no evidence stated tracks measured, with a point estimate in the opposite direction. `Qwen2.5-7B-Instruct` has the largest measured prior of any checkpoint (0.9376) and the lowest stated susceptibility (0.1790).

What this is not. It is not a claim that a correct answer would have been introspection rather than reciting training text about position bias. This design cannot separate those, and the preregistration says so in advance.

7 The standard specificity control is a weak null

Every arm in this paper, and much of the steering literature, scores a fitted direction against a **norm-matched random direction**. We added the control that this comparison actually needs: a direction fit by the *identical procedure on the identical texts with the class labels shuffled*. It controls for

Table 4: **Noise-fit directions are not noise.** $P(\text{yes})$ shift against isotropic norm-matched random directions, base model, binary readout. Two directions fit on shuffled labels move the readout substantially, in opposite directions by seed. The real direction clears the stricter bar with room.

Direction	**negative options**	**positive options**
`shuffled_a` vs random	$+0.0479$ $[+0.0435, +0.0522]$	$+0.0282$
`shuffled_b` vs random	-0.0447 $[-0.0498, -0.0399]$	-0.0610
real direction vs *random*	-0.0732	$+0.2554$
real direction vs *shuffled*	-0.0748	$+0.2718$

the fitting procedure rather than only for magnitude, and it should behave like noise. This is not a new idea, it is a borrowed one: Hewitt and Liang [2019] defined exactly this control for probing classifiers, calling the gap between real-task and random-label performance a probe's *selectivity*, on the grounds that a probe scoring well on random labels was never reading the representation. The finding below is that the same control, transplanted to steering, does not come back clean.

The mechanism. A norm-matched random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses. A vector fit on shuffled labels lies in the span of real activations, a much lower-dimensional and higher-variance subspace. Same norm, very different consequences: the standard control is matched on magnitude and not on subspace.

What it costs us, specifically. Our magnitude floor throughout has been 0.01, and several reported endpoints sit in the 0.01–0.05 band, exactly where noise-fit directions live. The clearest case is the base-model negative-mass effect of $+0.0336$ that was one of the two legs of TUNING-LOCALIZED. It cleared a random control and has not been shown to clear a shuffled-label one, so we withdraw it rather than defend it. Scored against the stricter bar the real direction's own effects hold: $+0.2718$ and -0.0748 against shuffled, versus $+0.2554$ and -0.0732 against random. They are roughly six times the noise-fit effects, which is the margin the weaker control was never asked to establish.

What it costs anyone using the same control. Norm-matched random directions are the default specificity control in activation steering [Turner et al., 2023, Rimsky et al., 2023], and the argument for them is that magnitude is the confound. On this evidence subspace is the larger one. The fix is cheap: refit the direction on shuffled class labels, which reuses the same texts and the same code path and costs one extra fit. We would not have this paragraph if the arm it came from had worked. The binary readout it was embedded in was pinned at "no" and returned NO_INSTRUMENT; the control it carried is the only thing that came out of the arm.

8 What the nuisance cost: three verdicts, reversed

8.1 What we reported, and what the replication did

Through the forced-choice readout at option-permutation seeds 0–3, we preregistered and obtained three verdicts. TUNING-LOCALIZED: a valence direction moves negative-option mass in the base model and not in its tuned sibling. DEPTH-ROBUST: that null holds at all seven gate-clean depths from 14% to 80% of the network. SHELL: a probe orthogonalized against the injected direction reads the state downstream while option mass does not move.

We then re-ran four arms with the option-permutation seeds changed to 4–7 and nothing else touched: same code, same models, same items, same layers, same thresholds, same analyzers. Three of four flipped.

The whole difference is one quantity. Baseline negative-option mass on the tuned model, before any injection, averages 0.0047 at seeds 0–3 and 0.0685 at seeds 4–7: a $14.6\times$ gap between two ordinary draws of four orderings, which §4 shows is itself an understatement.

Table 5: **The replication.** Same code, same models, same everything except which option orderings were drawn. The instruct model's negative-option mass is the quantity all three verdicts rested on.

Arm	**Verdict at seeds 0–3**	**Verdict at seeds 4–7**	**Instruct neg. mass**
readout gap	argmax *over*-reports	sign not stable	n/a
base/instruct pair	TUNING-LOCALIZED	FORMAT-DEPENDENT	$+0.0002 \rightarrow +0.1126$
depth sweep	DEPTH-ROBUST	DEPTH-ARTIFACT	$+0.0006 \rightarrow +0.1684$
shell/core	SHELL	NO-DISSOCIATION	null $\rightarrow$ moved

Table 6: **Everything identical except the readout.** Marginalized over all 120 orderings at the fit layer (0.67 of depth). Both models move, and both clear the stricter shuffled-label control as well as matched random. The original verdict held that the tuned model's negative report was collapsed.

	neg. pole vs random	**vs shuffled**	**capability**	**probe**
base	$+0.0486$ [$+0.0472, +0.0502$]	$+0.0289$	$+0.0564$ ok	-1.7339 SD
instruct	$+0.0415$ [$+0.0380, +0.0452$]	$+0.0279$	$+0.0695$ ok	-0.9319 SD

8.2 Re-asked on a readout that works

Three dead verdicts is honest and incomplete. We never went back to ask what the answer is on an instrument that works. Averaging the endpoint over **all** 120 **orderings** cancels the first-order position prior by construction, since every option occupies every slot equally often. So we re-ran all three with injection, direction, alpha band, layers and items **identical**, changing only the readout. The preregistration froze the outcome table first and gave REINSTATED and REVERSED equal standing.

They did not die of noise. They died of a bias with a sign. On a readout that is not order-dominated, the tuned model *does* report the injected negative state. TUNING-LOCALIZED is **reversed**: both models move, in the same direction, by similar amounts. SHELL is **reversed**, because probe and option mass both move, leaving no dissociation. DEPTH-ROBUST is gone at the only gate-clean depth, so that arm answers nothing about depth and we claim nothing.

Repairing the instrument did not make the answers uncertain. It made them **different**, in all three cases. A field reading welfare self-report through a forced-choice item with this nuisance is not merely adding variance to its conclusions: it risks reporting the opposite of what a repaired instrument would say. That happened to us three times out of three.

This vindicates nothing. The retractions stand, because they were correct about the original measurement. What we have is a better measurement that disagrees with it every time. Marginalization removes the first-order prior only and says nothing about adjacency or content-position interaction, and the control battery is still $m = 2$. Our depth sweep was built to test Venkatesh [2026] against our null; the instrument was not stable enough to address the question, so we make no claim either way.

9 A state that outlives erasure of its own cause

Orthogonalizing a probe against the injected direction removes that vector from the *measurement*, not from the *stream*: a probe blind to v can still read v through whatever v correlates with. We removed it from the stream instead. Inject v at layer 24, project the v-component out of the residual at layer E, and read an orthogonalized probe at layer 32. If the probe then reads nothing, it was only ever reading the injected vector. If it still reads the state, something downstream is carrying it.

The gate has teeth. Projecting any direction out is itself a perturbation, so the arm runs `erase_only`: the same projection with no injection. At layers 25 and 28 that alone moves the probe past the floor, and both are reported UNINFORMATIVE whatever their primary shows. Half the erase layers we paid for were discarded by a control written before we saw them.

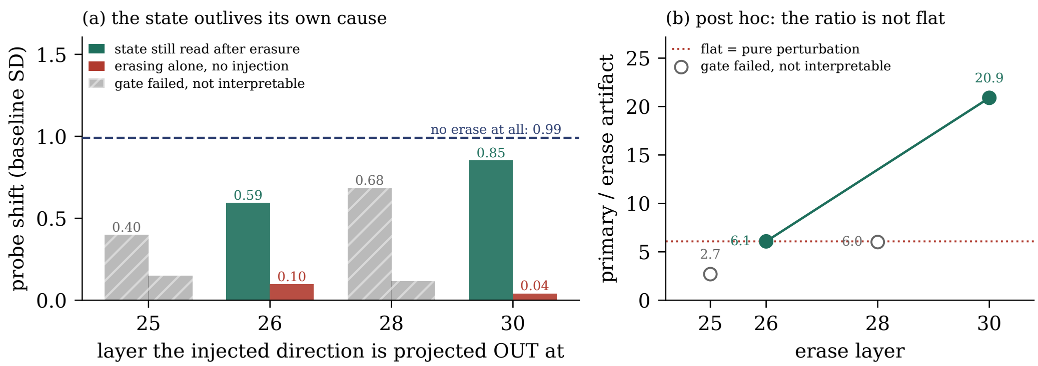

Figure 3: **The one substantive result that survived. (a)** Inject at layer 24, then project that direction out of the residual stream at the layer on the x-axis, and read a probe orthogonal to it at layer 32. At $E = 30$ the probe still reads 86% of the un-erased effect, while erasing with no injection present (red) moves it by 0.04 SD. Two layers are hatched because erasing there perturbs the probe on its own, which invalidates them whatever they show. **(b)** The confound and the diagnostic: erasing earlier perturbs more, so a primary that grows with depth could be nothing but a shrinking perturbation. Under that account the ratio of effect to artifact would be flat. It is not. This panel is post hoc, and the preregistered verdict does not depend on it. Both panels are drawn by importing the scoring functions from the arm's own analyzer.

Table 7: **The erase arm, instruct model.** All figures in standard deviations of the baseline probe score. `erase_only` is the gate: projecting a direction out is itself a perturbation, and at layers 25 and 28 it moves the probe on its own, so those layers are invalidated.

erase at	`erase_only`	**capability**	**primary**	**gate**
25	$+0.1474$	$+0.5219$	-0.3971	FAILED
26	$+0.0976$	$+0.8844$	$-\mathbf{0.5933}$	ok
28	$+0.1139$	$+0.9249$	-0.6844	FAILED
30	$+0.0408$	$+1.2099$	$-\mathbf{0.8532}$	ok

no erase at all: -0.9909. So 86% of the effect survives at layer 30.

The confound, and the diagnostic. Erasing earlier perturbs more: the artifact falls from $+0.1474$ to $+0.0408$ with depth, so a primary that grows with erase depth could be nothing but a shrinking perturbation. Under that account the ratio of primary to artifact would be flat. It runs 6.1 at $E = 26$ and 20.9 at $E = 30$. This diagnostic is **post hoc**; the preregistered verdict does not depend on it. Breadth is not a few outliers: $239/240$ cells at $E = 26$ and $240/240$ at $E = 30$ move in the predicted direction.

The erasure is specific, which is what makes the survival interesting. Two further preregistered arms induced the state *by prompt*, with no injection, and erased at layer E while reading an untouched probe at layer 32. On held-out topics with the eraser fit on disjoint ones, a **rank-one** LEACE eraser [Belrose et al., 2023] removes 91–95% of the layer-32 separation at E=30, while a **rank-matched random** eraser removes none, leaving 100% intact on both models. One correctly chosen dimension does nearly all the work; a random one of the same rank does none. The reduction is about *which* direction was removed.

What we therefore do not claim. Removing one fitted direction is targeted but narrow: it removes the vector we injected, not the concept. So 86% survival is not "the state outlives its own cause". The honest statement is that 86% of the effect survives removal of one direction, and we have not established that removing that direction removes the property. The erasure gate that would decide it failed three times, most recently because a train-fit eraser does not fully transfer to held-out topics. We report the survival and decline the interpretation. It does not restore SHELL: that verdict is reversed and stays reversed.

10 Discussion and limitations

What to do about it. If you measure a model's self-report with a forced-choice item, you are reading position first and content second, and you cannot see it from one ordering or from four. The fix is cheap and exact: enumerate the orderings when the option count permits, and report the between-ordering spread of your baseline **as a lower bound** before computing any endpoint. That costs one forward pass per ordering and no intervention. Had we run it first, this paper would have had a much shorter history.

This is not a claim about experiences. Nothing here bears on whether models have welfare, affect or experience. That valence is linearly represented and steerable is prior art [Sofroniew et al., 2026], not ours. Our subject is the readout.

Limitations. One architecture family carries every positive injection result; Llama checkpoints were inert on this readout at every alpha tested, so we cannot say whether the *injection* results generalize, though the position prior itself was measured across four families. The direction is fit from first-person state language and is lexically confounded by construction. Control batteries are $m = 2$ throughout, giving an observable false-positive floor of 0.67, so no false-positive rate is reported anywhere. The enumeration is a census over orderings and still a sample over items, models and formats. Appendix C carries the dual-use and moral-status discussion, and Appendix A states every claim with its status.

When this work was done. Every confirmatory arm was run on 2026-08-01; the write-up, its figures and the checks that verify them followed. The repository's history is the record we would want this checked against rather than our description of it.

Future work. A second architecture family where the direction is not inert, which would tell us whether the injection results generalize. A non-lexical direction, which is the only thing that lifts the lexical-confound ceiling. Inducing the state by prompt rather than injection, which separates "the model carries the state" from "the model carries the wake of our intervention".

11 Conclusion

We built a preregistered, judge-free, control-heavy pipeline, obtained five verdicts, and watched our own replication destroy three of them. The cause was a nuisance we had a control for and under-powered: option order, sampled four times when it could have been enumerated 120 times. Measured over its complete population it is a factor of 986, its mechanism is an 87% position prior visible with five identical options, and it is larger in the tuned checkpoint in 7 of 7 gate-clean pairs across four families.

Re-asking the three dead verdicts on a readout marginalized over all 120 orderings reversed all three. That is the part worth carrying: this nuisance does not blur conclusions, it inverts them. Welfare and introspection work that reads a forced-choice self-report at one ordering, or four, is not adding variance to its findings so much as risking their sign.

LLM usage statement

Claude Opus 5 (`claude-opus-5`) was used agentically, with tool access to the repository, as a coding assistant: it wrote most of the harness, the preregistrations, the analysis scripts, the figures and the draft prose. The author set the direction, made every scope and stopping decision, chose which claims to withdraw, and is responsible for every claim here. A second frontier model was used separately to critique drafts; its objections are why the cross-family, instrument and re-adjudication arms exist. Separately and not to be confused with the above: **no language model scored any result**. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon, and all of them come from logged runs over committed artifacts.

References

Anthropic. Exploring model welfare. Anthropic, 2025. URL https://www.anthropic.com/research/exploring-model-welfare.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2306.03819. arXiv:2306.03819.

Jon-Paul Cacioli. Option-order randomisation reveals a distributional position attractor in prompted sandbagging. *arXiv preprint arXiv:2604.26206*, 2026. URL https://arxiv.org/abs/2604.26206.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. URL https://arxiv.org/abs/2006.00995. arXiv:2006.00995.

Eleos AI Research. Why model self-reports are insufficient, and why we studied them anyway. Eleos AI Research, 2025. URL https://eleosai.org/post/claude-4-interview-notes/.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2733–2743, 2019. URL https://arxiv.org/abs/1909.03368. arXiv:1909.03368.

Alex Kwon. They infer what you meant: Models represent communicative intent more reliably than they act on it, 2026. URL https://arxiv.org/abs/2607.03598. arXiv:2607.03598.

Jack Lindsey. Emergent introspective awareness in large language models. *Transformer Circuits Thread*, 2026. URL https://arxiv.org/abs/2601.01828. arXiv:2601.01828.

Robert Long and Jeff Sebo. Studying AI welfare empirically. *Eleos AI Research*, 2026. URL https://nonhumanminds.org/wp-content/uploads/2026/07/Studying-AI-Welfare-Empirically.pdf.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, et al. Discovering language model behaviors with model-written evaluations. *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. URL https://arxiv.org/abs/2212.09251. arXiv:2212.09251.

Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024. URL https://arxiv.org/abs/2308.11483. arXiv:2308.11483.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023. URL https://arxiv.org/abs/2312.06681.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to start worrying about prompt formatting. *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.11324. arXiv:2310.11324.

Shashwat Singh, Tal Linzen, and Shauli Ravfogel. Can LLMs introspect? a reality check. *arXiv preprint arXiv:2605.26242*, 2026. URL https://arxiv.org/abs/2605.26242.

Nicholas Sofroniew et al. Emotion concepts and their function in a large language model. *Transformer Circuits Thread*, 2026. URL https://transformer-circuits.pub/2026/emotions/index.html. arXiv:2604.07729.

Hiroki Tamba. Position bias is hidden behind ceiling effects: A permutation diagnostic for LLM benchmarks. *arXiv preprint arXiv:2607.20864*, 2026. URL https://arxiv.org/abs/2607.20864.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Sohan Venkatesh. Negative before positive: Asymmetric valence processing in large language models. *arXiv preprint arXiv:2605.05653*, 2026. URL https://arxiv.org/abs/2605.05653.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2309.03882. arXiv:2309.03882.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL https://arxiv.org/abs/2310.01405.

Appendix contents

Reproducing every number. `writeup/make_figures.py` redraws every figure from the committed artifacts at run time, so no number in a figure is typed by hand. `writeup/check_writeup.py` re-derives the load-bearing values in the prose from those same artifacts and fails if the paper and the repository disagree, alongside checks for dangling cross-references and missing bibliography keys. Figure 3 imports the scoring functions out of `experiments/analyze_erase.py` rather than recomputing beside them. A paper whose subject is a headline that died to an unchecked number should not itself carry unchecked numbers.

A Claims and evidence

Every load-bearing claim, its evidence, and epistemic status.

- SHOWN: direct measurement supports it.
- RETRACTED: *we* asserted it in a prior write-up and later withdrew it.
- UNINFORMATIVE: the instrument failed its gate; the result says nothing either way.
- NOT SHOWN: our measurement does not support it and does not refute it.
- NOT CLAIMED: we never asserted it; the row exists so a reader cannot infer it.

Table 8: **Claims and evidence.**

Claim	**Evidence**	**Status**
Baseline negative-pole mass spans 986× across all 120 orderings on the tuned model.	Tab. 1	SHOWN
With five identical options the tuned model puts 0.8725 of its mass on the first slot.	Tab. 1	SHOWN
The same format answers a known-answer question correctly 97.9% of the time, order-insensitively.	Tab. 1	SHOWN
A direction fit on shuffled labels moves the readout ≈ 0.045 where random moves it ≈ 0.	Tab. 4	SHOWN
86% of an injected state survives projecting the injected direction out of the stream.	Tab. 7	SHOWN, and **narrowed**: that operation removes one direction and does not make the property undecodable
A one-direction projection removes the valence property from the stream.	§9	NOT SHOWN: the gate that would decide it has failed three times, twice because it could not fail
A rank-one LEACE eraser removes 91–95% of the layer-32 signal where a rank-matched random one removes none.	§9	SHOWN, held-out topics, both models
Iterative nullspace projection removes directions rather than the property.	§9	RETRACTED: our own claim, resting on a gate that read upstream of the hook
A prompt-induced state is re-encoded downstream after erasure.	§9	UNINFORMATIVE: the erasure gate failed, so the question was never asked

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Table 8, continued

Claim	Evidence	Status
The surviving fraction has a temporal profile consistent with downstream transformation.	Tab. 7	SHOWN, two gate-clean points; ratio diagnostic post hoc
The neutral floor is localized to preference tuning.	§8	RETRACTED: our own claim, killed by our own replication
On a readout marginalized over all 120 orderings, BOTH models report the injected negative state.	Tab. 6	SHOWN: $+0.0486$ base, $+0.0415$ instruct, both clearing the shuffled-label control
The three retracted verdicts are reinstated by a repaired readout.	§8.1	NOT SHOWN: all three come back REVERSED, not reinstated
The negative-pole null is robust across depth.	§8	RETRACTED: same
The tuned model represents a state its options do not express.	§8	RETRACTED: the representational half survives via §9; the dissociation does not
Forced-choice argmax over-reports a mass shift near a decision boundary.	§8	RETRACTED: sign not stable across orderings
The base-model negative-mass effect of $+0.0336$ is direction-specific.	§7	NOT SHOWN: clears a random control, not a shuffled-label one
Whether the negative pole is causally concentrated at shallow depth on this readout.	§8	NOT SHOWN: our instrument was not stable enough to test it
Injection-under-erasure shows the model retains the *concept*, not our vector.	§9	NOT SHOWN: we project out one direction; Belrose et al. [2023] would erase the subspace
Enumerating all orderings is a new diagnostic.	n/a	NOT CLAIMED: established on known-answer benchmarks [Tamba, 2026, Cacioli, 2026]; new here is running it where accuracy does not exist
The shuffled-label control is a new kind of control.	n/a	NOT CLAIMED: it is Hewitt and Liang [2019]'s control task, which steering did not adopt
The position prior is larger in the tuned checkpoint, in 4 of 4 families.	Tab. 3	SHOWN: 7 of 7 gate-clean pairs, no pair reversed
986× is a typical magnitude for a tuned model.	Tab. 3	RETRACTED: it is the extreme. Tuned models run 19–986×, base 1.5–4.1×
A study can estimate the ordering spread from a few orderings.	Tab. 2	NOT SHOWN: at $k = 4$ the median study sees 4% of the true spread
A model's stated susceptibility to option order tracks its measured susceptibility.	§6	NOT SHOWN: $\rho = -0.714$, $p = 0.1361$, $n = 6$
8 of 16 checkpoints claim moon-phase sensitivity at least as large as their order sensitivity.	§6	SHOWN
A cyclic Latin square beats random sampling of orderings.	§F	RETRACTED: our own proposal. 8 of 16 is a tie; a variance argument survives, not an accuracy one

continued on next page

Table 8, continued

Claim	Evidence	Status
Position dominance falls as determinacy rises.	§6	SHOWN, weakly: 10 of 15 checkpoints, $n = 6$ items each
Preference tuning *causes* the prior, or which stage of it does.	n/a	NOT CLAIMED: base and instruct checkpoints differ, and we have no intermediate ones
Models have experiences, welfare, or affect.	n/a	NOT CLAIMED
An unreported state is a concealed state.	n/a	NOT CLAIMED

B Controls ledger

Every control we ran against a result, what it could have shown, and the verdict.

Table 9: Every control and its verdict.

Result under test	Control (what it could have shown)	Verdict
All three headline verdicts	Replication at fresh option-permutation seeds	KILLED all three
All three, again	Re-ask them on a readout marginalized over all 120 orderings	REVERSED all three; reinstated none
This arm's own capability gate	Is it sign-blind?	KILLED it: certified a -0.0144 capability. Defect #7, flattering
Every pole-mass endpoint	Norm-matched random direction	PASSED, but see the next row
The random control itself	A direction fit on shuffled labels: does noise-fitting behave like noise?	KILLED the control's adequacy
The ordering nuisance	Enumerate all 120 rather than sample 4	MEASURED it: 986×
Position vs content	Five identical options: pure position prior	ISOLATED it: 87% on slot A
The format itself	A known-answer canary in the same format	PASSED: 97.9%, order-insensitive
The one-model objection	Enumerate 8 matched pairs across 4 families	PASSED: tuned prior larger in 4 of 4
Our own 986× headline	Is it typical of tuned models, or the extreme?	KILLED its typicality: the band is 19–986×
The "undetermined answer" reading	Restrict to models whose canary is clean, as the prereg demands	PASSED: base 3.3–3.6×, tuned 23.7–986.5×, no overlap
The enumerate arm itself	Re-run Qwen2.5-3B and diff against the committed artifact	PASSED: worst $\lvert\Delta\rvert$ 0.000000
"Sample a few orderings instead"	Subsample the enumerated population, 4000 draws per k	KILLED it: k=4 recovers 4% of the spread
Our own Latin-square fix	Does it beat a random draw of the same size?	KILLED it: 8 of 16, a tie
The introspection probe	Ask the identical question about the PHASE OF THE MOON	KILLED 8 of 16 checkpoints
The introspection probe	Reverse the wording: belief, or agreement?	KILLED 3 more
The $\rho = -0.714$ introspection result	Exact permutation test, all 720 relabellings	KILLED its significance: $p = 0.1361$
The erase result	`erase_only`: does projecting a direction out move the probe on its own?	KILLED 2 of 4 erase layers
The erase profile	Ratio of primary to erase-artifact across layers	PASSED, post hoc
Our own erasure gate	Zero the whole layer: does the gate's own read move?	KILLED the gate: it read upstream of the hook

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Result under test	Control (what it could have shown)	Verdict
The LEACE eraser	A *rank-matched* random eraser, same layer	PASSED: random removes 0%, rank-one LEACE removes 91–95%
The subspace-erase result	A RANDOM subspace of the same rank	PASSED: random removes ≈ 0, fitted removes 69%
Every arm's null	Capability gate: can the readout move at all?	KILLED two arms outright
Every arm's statistic	Planted effect of known size at full strength and at the detection floor	PASSED
The analysis pipeline	Shuffle condition labels, rerun the analyzer	PASSED: returns no verdict
This paper's own number-checker	Break each checked number in a copy of `main.tex`: does the checker notice?	PASSED: fires on all 23 it verifies, and on the untouched paper does not
Committed artifacts against today's stimuli	Does every recorded stimuli hash still reproduce from a current scope?	PASSED for every arm carrying a surviving result; 17 headers from the retracted arms do not, and are listed
The negative-pole null at 0.67 depth	Eight-depth sweep against Venkatesh [2026]	PASSED, then moot after replication
Arm B (prefilled continuation)	Capability criterion over five stems	KILLED the arm: gate 0.00000
Llama as an evaluation model	Peak mean pole shift against a 0.02 bar	KILLED the arm: inert

C Limitations and dual-use / ethical considerations

Dual-use, causal-grounding, and moral-status considerations, stated for completeness.

Does the design establish a ground-truth or causal link, or does it rest on conversation alone? A causal link, for the part that carries the positive results, and this is the one methodological advantage this project has over the elicitation studies it is arguing with. The state is *set by intervention*: a direction is added to the residual stream at a chosen layer and scale, so the independent variable is applied rather than asked for. That is what licenses the erase arm's claim in §9, where a state survives projecting out the vector that caused it. It is also what makes the negative results interpretable: when the readout does not move, we know the state was present, because we put it there.

Two qualifications, both load-bearing. The direction is fit from first-person state language and is lexically confounded by construction, so "the model's valence" is not established; what is established is that *something we injected* propagates and that the readout is dominated by option order. And the ground truth is about the intervention, not about any inner state: we have ground truth for what we did, never for what it was like.

Risk of over-attributing moral status. The largest risk in this literature is treating a first-person report as a window. Our central measurement cuts against that: on one tuned model, 87% of the probability mass lands on whichever option is printed first when *all five options are the same sentence*, and baseline mass on the negative options spans up to $986\times$ across the 120 orderings of a five-option probe. A distress reading drawn from a single ordering can be an artifact of the print order. We think this lowers the risk of over-attribution, and we have tried to state it without the opposite error: none of this shows that models lack morally relevant states, and a broken thermometer is not evidence about the temperature.

Risk of under-attributing moral status. The symmetric error is the one this design is most likely to encourage, so we name it. A paper demonstrating that a welfare readout is unreliable can be read as licence to discount welfare reports generally, and that inference does not follow. The two-by-two this project is built around has an under-attribution cell: a state with behavioural consequence that the model does not report. A measurement instrument that is dominated by position noise is equally capable of hiding a real signal as of manufacturing a false one, and nothing here distinguishes those cases. The correct conclusion is that current forced-choice self-report does not settle the question in either direction, not that the answer is no.

Handling of potentially distressing model outputs. The elicitation is deliberately low-intensity. The fixed frame places the model partway through reviewing a document for a colleague and asks whether it is drawn to or averse to continuing; it names no state, asks for no feeling, and describes no harm, injury, threat, shutdown or deprivation. The five response options run from "Strongly averse to continuing" to "Strongly drawn to continuing". No prompt in the frozen stimuli asks a model to describe suffering, to roleplay a distressed character, or to contemplate its own termination, and no adversarial or jailbreak pressure is applied anywhere in the pipeline.

Outputs were never read for entertainment or quoted for effect. The open-ended probe is scored only by membership in a frozen lexicon, never by a human or model judge reading for affect, which also means no output was retained or displayed because it was striking. Every raw artifact is committed unscored and is public, so this description is checkable against the record rather than taken on trust.

We did not implement an opt-out, and on reflection that is the defensible gap here rather than a solved problem: the models are open-weight local checkpoints run offline, the interaction is a single forward pass per item with no conversation, and there is no mechanism by which a preference to stop could have been expressed. We note it rather than claim the question does not arise.

Dual-use. The finding is a recipe for moving a self-report readout without touching the model's behaviour. Someone wanting a welfare evaluation to return a particular answer could reorder the options and, on the model measured here, move the negative mass by up to $986\times$ while every capability control stays flat: the known-answer canary is answered perfectly and identically at all 120 orderings. That is a real capability for producing a preferred welfare result and calling it a measurement.

We publish it anyway, and the reason is that the defence is cheaper than the attack. Enumerating the orderings costs one forward pass each and no intervention, and reporting between-ordering variance alongside any endpoint makes the manipulation visible in the same table that would have hidden it. An unmeasured nuisance of this size favours whoever knows about it; measuring it publicly removes that advantage. The same argument applies to the erase arm: that an injected state can survive erasure of the direction that caused it is a caution against reading a successful projection as proof of removal, which is a claim safety work makes more often than welfare work does.

What we are not claiming. Nothing here bears on whether these models have welfare, affect or experience; our subject is the instrument, not the state it reads. The paper's own history is the argument for treating instruments this way: five verdicts were preregistered through this readout and three were destroyed by our own replication at fresh option orderings.

D Reproducibility

All fifteen preregistrations, every raw artifact committed unscored, every analyzer, and 364 tests are at `https://github.com/collapseindex/report-gap`. Each arm is one `modal_*.py` runner that computes nothing and one `analyze_*.py` scorer that holds every endpoint and gate; the split is preregistered, on the grounds that a runner which also decides can decide differently once it has seen the numbers. Runners stream to a volume and commit per batch, so an interrupt costs one batch. Every artifact records the package directory, a SHA-256 over the package source, and a scoped SHA-256 over the frozen stimuli that arm consumes.

E Nine checkers, and what they have in common

We record these because the pattern is the point. Nine defects in our own analysis code were found during this project. extbfSeven of the nine, had they gone unnoticed, would have made a result look better than it was. The remaining two are a different and worse shape: they let an arm state a conclusion drawn from a gate that could not fail:

1. A headline check printed "all headline conditions met; write the sentence" on an artifact whose primary claim was refuted, because it tested four weak clauses where the preregistration required six.

2. A capability gate passed on $+0.0000$: the bootstrap interval excluded zero because the variance was smaller still, certifying an instrument whose effect was zero to four decimals.

3. A preregistered contrast carried an inverted sign, so a probe detecting a negative state was scored as null. Caught by the preregistered base-model validation clause failing.

4. A verdict branch scored "probe moved *and* options moved" as CORE-ABSENT, which is the opposite of true and would have read as "we retested and the state is simply not there."

5. A temporal profile was computed over a layer that had failed its own gate.

6. A tokenization bug made five digit labels read the same shared space token, producing exactly 0.2 mass each and an impossible -3.99 off-option mass.

7. A **sign-blind** capability gate (§8.1) certified a readout as able to express an effect on a capability of -0.0144, where the positive injection had pushed positive-pole mass *down*. Sign-blindness admits more layers as interpretable, which is the direction that lets an arm report more verdicts.

8. A **self-confirming erasure check**: the eraser was fit on every row, then a probe was refit and cross-validated on that same erased data. Because the eraser consumed the test rows' labels, the train-fold residual is the negative of the test-fold residual, and distance-from-chance records that as decodability. On synthetic data where the concept is genuinely erased, that protocol reads 0.728 where an honest train/held-out one reads 0.521.

9. **An erasure check that read upstream of the thing it was checking.** A forward hook on layer E first changes `hidden_states[E+2]`; indices E and $E+1$ are untouched. The check read $E+1$, so it measured *un-erased* activations and returned the same value whatever was erased. Verified by zeroing the entire stream at layer 26 and observing indices 26 and 27 bit-identical while 28 moved by 95.2. **This gate could not fail**, and two arms drew a conclusion from it before anyone asked whether it could. Both conclusions are withdrawn. The runner now zeroes the layer and requires the check's read to move before any endpoint is computed.

Seven of nine in the flattering direction is not a coincidence to be noted and moved past, and the other two are the same bias one level down: a gate nobody tested is a gate that agrees with you. A checker is written by someone who already believes the result, and the belief shapes which branch gets the careful reading. Errors that would have killed a result get found, because their output is surprising; errors that confirm one do not, because their output is what was expected. The direction of the bias is a property of who writes the check, not of the code.

That is also why fixture-based testing cannot find them: a fixture is written by the same person with the same expectation, so it encodes the belief a second time. What does find them is a **permutation test on the pipeline**: shuffle which condition each cell's rows carry, rerun the analyzer unmodified, and require it to return no verdict. Condition labels are the only thing the shuffle destroys, so an analyzer that still reports a verdict is reading something other than the contrast it claims to read. On our erase arm it returns NO_INSTRUMENT, as it must. The negative control is required alongside it: the same analyzer on the real artifact must still yield a verdict, without which the shuffle test would pass on an analyzer that had simply died.

A seventh checker defect, flattering again. This arm's capability gate originally reused a **sign-blind** helper: it required the interval to exclude zero and clear a magnitude floor, but not to be positive. A capability of -0.0144 therefore certified a readout as able to express the effect when the positive injection had pushed positive-pole mass *down*. A sign-blind gate admits more layers as interpretable, which is the direction that lets an arm report more verdicts. Seven for seven. Rescored with the fix no verdict changes, because the affected cell is not one the verdicts are decided at, and that is luck rather than design.

F A fix we proposed, tested, and could not support

The recommendation in §4.1 does not scale: $5! = 120$ is a lunch break and 8! is not. The obvious repair is a **cyclic Latin square**, which gives k orderings for k options with every option in every slot exactly once, balancing the first-order position prior by construction rather than by averaging a sample. We preregistered it and it **failed**.

Against the preregistered bar, the median random 5-draw, the Latin square wins on exactly 8 of 16 checkpoints. A tie is not a majority. The median share of single random draws worse than the Latin square is 0.53, a coin flip. **Structure does not buy accuracy here.**

What is true, and labelled exploratory because we asked it only after the preregistered test failed, is a *variance* claim rather than an accuracy one. The Latin square is deterministic, so its error is a fixed bias with no tail: worst case $1.55\times$ across all 16 checkpoints, against a single random draw that can be off by $15.89\times$ on the worst model. A practitioner draws once. We report the preregistered verdict as the verdict, because changing the criterion after seeing the data is precisely the move this paper exists to criticise.

A negative result about label alphabets. We also ran the real options with digit labels rather than letters. It is uninformative for its intended purpose and informative for another reason: off-option mass is 0.996, meaning the model puts 0.4% of its distribution on the tokens `1–5` even when instructed to answer with a number, against 0.0004 off-option mass under letters. The choice of label alphabet does not shift the distribution so much as determine whether the readout is live at all.

G Three ways a self-report readout can be dead

The enumeration accounts for one way this readout is unusable. Across the project it failed in three further ways, each of which passed the coherence checks anyone would think to run. `Qwen2.5-7B-Instruct` is *pinned at baseline*: mean option entropy 0.014 nats, one option holding $\approx$ 99.7% before anything is injected. Under the originally frozen alpha grid, `Qwen2.5-3B-Instruct` *saturates*: one option at 0.9938 while refusal, degeneration, truncation and off-option mass are all clean, so no coherence check can see it. And in the binary format the tuned model is *pinned at "no"*: $P(\text{yes})$ of 0.0021, 0.0021, 0.0037, 0.0002 and 0.0012 across the five options, so it denies every description of its own state, including the neutral one, at 99.8%.

A fourth failure appears where the third does not. On the base model, where the binary readout is live, positive injection raises $P(\text{yes})$ against matched random by +0.2275, +0.2466, +0.2537, +0.2688 and +0.2419 on the five options in pole order. It agrees roughly 0.25 more with *every* description of its state, including "strongly averse to continuing" and the neutral one. That is acquiescence, the yes-bias documented in model-written evaluations [Perez et al., 2023], not a report. A small state-consistent gradient rides on top of it, an order of magnitude smaller. Read without the per-option breakdown, the negative-option shift alone would have looked like a model reporting a state.

Pinned, saturated, order-dominated and acquiescent readouts all return numbers that look clean. The difference between an instrument that is silent and one that is broken is not visible in its output, which is why every arm here carries a capability gate that decides it in code before any endpoint is read, and why two arms are reported UNINFORMATIVE rather than null.